Overview of ESDD2: Environment-Aware Speech and Sound Deepfake Detection Challenge

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***Abstract*—Audio recorded in real-world environments often contains a mixture of components: foreground speech and background environmental sounds. With rapid advances in text-to-speech, voice conversion, and other generation models, either component can now be modified independently. Such component-level manipulations are harder to detect, as the remaining unaltered component can mislead the systems designed for whole deepfake audio, and they often sound more natural to human listeners. To address this gap, we are launching the second Environment-Aware Speech and Sound Deepfake Detection Challenge (ESDD2), which focuses on defending against more challenging and realistic environment-aware, component-level deepfake manipulations. ESDD2 is built on our recently released CompSpoofV2 dataset and uses a separation-enhanced joint learning framework as the baseline. This paper presents an overview of the ESDD2 Challenge: the task description, the new database, baseline, metric, and a summary of the results.**

***Index Terms*—audio sound deepfake detection, audio anti-spoofing, component-level audio anti-spoofing**

I. Introduction

In real-world recording scenarios, audio often contains a mixture of foreground speech and background sounds. We define such audio as comprising two components: (i) speech, i.e., linguistically meaningful speech produced by the primary foreground speaker, and (ii) environmental sound, i.e., any non-speech background or non-target speech. With rapid advances in text-to-speech, voice conversion, and other generation models, either component can now be modified independently. For example, replacing the background while leaving the foreground human speech unchanged, or modifying the speech content while preserving the background. Such component-level manipulations are harder to detect, as the remaining unaltered component can mislead the systems designed for whole deepfake audio, and they often sound more natural to human listeners.

To address this gap, we introduced a component-level spoofing audio dataset CompSpoof [1] in 2025. Then, we extend the dataset to CompSpoofV2, which contains more than 250,000 audio clips (around 283 hours) formed by mixing bona fide (genuine or real) and spoofed (synthetic or manipulated) audio, including speech and environmental sounds, from multiple sources. We also proposed a separation-enhanced joint learning framework to address this gap.

To further promote the development of environment-aware speech and sound deepfake detection, we extend our first ESDD challenge [2]–[4] and launch the second Environment-Aware Speech and Sound Deepfake Detection Challenge (ESDD2) at ICME 2026. While our first ESDD challenge focuses on the detection of forged environmental sounds, ESDD2 advances this direction by addressing environment-aware component-level spoofing, where either or both speech and environmental sounds may be manipulated or synthesized, posing a more challenging and realistic detection scenario.

The main contributions are summarized as follows:

- We present a comprehensive description of the ESDD2 challenge, including the dataset, task description, baseline systems and evaluation criteria.
- We report and analyze the leaderboard results, providing a systematic comparison of participating systems.
- We summarize common effective design choices observed in top-performing systems, offering insights for future research on environment-aware speech and sound deepfake detection.

II. ESDD2 Challenge

A. Task Description

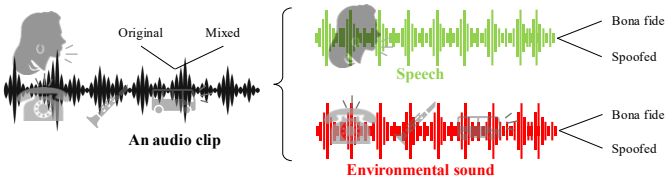


Fig. 1. ESDD2 Task illustration. An audio clip is first classified as mixed or original; for mixed audio, speech and environmental sound components are separately evaluated for genuineness, resulting in five target classes.

In ESDD2 and the discussed CompSpoofV2, we consider the scenario where each audio belongs to one of five categories, covering all combinations of bona fide and spoofed speech and environmental sound as shown in Table I. The task of this challenge is to classify a given audio clip into one of the five defined classes, as shown in Figure 1.

TABLE I
FIVE CATEGORIES CONSIDERED IN COMPSPOOFV2 DATASET.

ID	Mixed	Speech	Environment	Class Label	Description
0	✗	Bona fide	Bona fide	original	Original audio without any manipulation or mixing
1	✓	Bona fide	Bona fide	bonafide_bonafide	Bona fide speech mixed with bona fide environmental sound from other audio
2	✓	Spoofed	Bona fide	spoof_bonafide	Spoofed speech mixed with bona fide environmental sound
3	✓	Bona fide	Spoofed	bonafide_spoof	Bona fide speech mixed with spoofed environmental sound
4	✓	Spoofed	Spoofed	spoof_spoof	Spoofed speech mixed with spoofed environmental sound

TABLE II
AUDIO SOURCES FOR THE TRAINING AND VALIDATION SETS.

ID	Original Source	Speech Source	Environmental Sound Source
0	AudioCaps [5], VGGSound [6]	–	–
1	–	CommonVoice [7], LibriTTS [8], english-conversation-corpus [9]	AudioCaps, TAUUAS [10], TUTSED [11]–[13], UrbanSound [14], VGGSound
2	–	CommonVoice, LibriTTS	EnvSDD [15], VcapAV [16]
3	–	ASV5 [17], MLAAD [18]	AudioCaps, TAUUAS, TUTSED, UrbanSound, VGGSound
4	–	ASV5, MLAAD	EnvSDD, VcapAV

TABLE III
AUDIO SOURCES FOR THE EVALUATION AND TEST SETS.

ID	Original Source	Speech Source	Environmental Sound Source
0	AudioCaps, VGGSound	–	–
1	–	CommonVoice, LibriTTS, english-conversation-corpus	AudioCaps, TAUUAS, TUTSED, UrbanSound, VGGSound
2	–	CommonVoice, LibriTTS	EnvSDD, VcapAV, **New Generated by MeanAudio**
3	–	ASV5, MLAAD, **New Generated by SoulX-Podcast**	AudioCaps, TAUUAS, TUTSED, UrbanSound, VGGSound
4	–	ASV5, MLAAD, **New Generated by SoulX-Podcast**	EnvSDD, VcapAV, **New Generated by MeanAudio**

The training and validation sets share the same data sources and class distribution. Similarly, the evaluation and test sets share the same data sources and class distribution. Notably, the evaluation and test sets include newly generated audio samples that are unseen in the training and validation sets.

B. CompSpoofV2 Dataset

The **CompSpoofV2**[1] is a dataset designed for component-level anti-spoofing detection research, where either the speech or the environmental sound component (or both) may be spoofed. CompSpoofV2 contains over 250,000 audio samples, with a total duration of approximately 283 hours. Each audio sample has a fixed length of 4 seconds and is provided at multiple sampling rates, enabling a more faithful simulation of real-world acoustic and system-level variations. Building upon the CompSpoof dataset [1], CompSpoofV2 significantly expands the diversity of attack sources, environmental sounds, and mixing strategies. In addition, newly generated audio samples are distributed across the test set and are specifically designed to serve as detection data under unseen conditions. The audio sources for each category are summarized in Table II and Table III.

C. Baseline

We utilize a **separation-enhanced joint learning framework** [1] as the baseline for this task. As shown in Fig. 2, the framework consists of a mixture detection model, a UNet-based separation network, and two component-wise anti-spoofing models for speech and environmental sound. The framework first detects potentially spoofed mixtures. If the input audio is detected as containing spoofed mixtures, it will be separated into speech and environmental components. Subsequently, these components are processed by speech- and environment-specific anti-spoofing models. Their outputs are

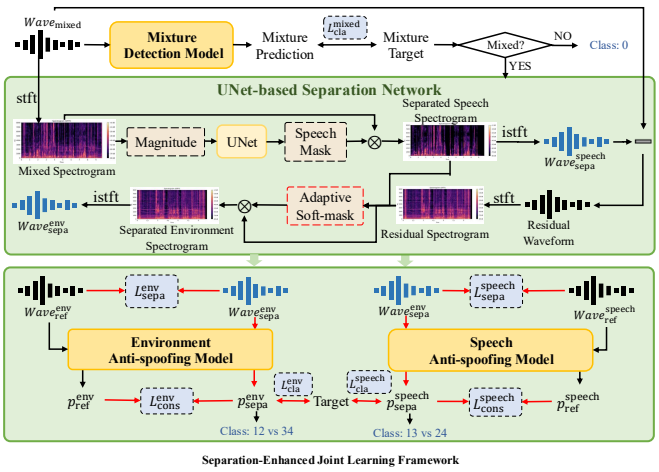


Fig. 2. Overview of the proposed separation-enhanced joint learning framework. '→' illustrates the joint learning data flow between the separation and anti-spoofing models.

[1]https://xuepingzhang.github.io/CompSpoof-V2-Dataset/

TABLE IV
PERFORMANCE OF THE TOP-12 SYSTEMS (F1-SCORE / ORIGINAL EER / SPEECH EER / ENVIRONMENTAL EER) "ENSEM" DENOTES THE NUMBER OF MODELS IN AN ENSEMBLE ("1" INDICATES A SINGLE MODEL).

Team No.	Team Affiliation	System	Ensem	Param.	Data Aug.	Eval set	Test set
1	AHU	E2E-EA-SSDD	7	6.56B	(1)(2)	-	0.8775 / - / - / -
2	CUC	EnvTriCascade	2	540.81M	(1)	-	0.8266 / - / - / -
3	SETW	FrozenSSL-Ens4	4	1908M	(1)(3)(4)(5)	0.8124 / - / - / -	0.8200 / - / - / -
4	HKUST(GZ)	GLADSE	2	674.57M	(1)	0.7120 / 0.0102 / 0.1201 / 0.0999	0.8077 / 0.0133 / 0.0896 / 0.0926
5	SIT	CompEnsFusion	8	4.6B	(1)	0.7715 / 0.0124 / 0.2177 / 0.1341	0.7828 / 0.0109 / 0.2164 / 0.1263
6	WHU	LaMSep-DF	1	356.85M	(1)	0.7045 / 0.0227 / 0.2889 / 0.2218	0.7262 / 0.0150 / 0.1171 / 0.0869
7	JAIST_1	MIF	4	1034.15M	-	0.6995 / 0.0560 / 0.1852 / 0.1436	0.7187 / 0.0345 / 0.1562 / 0.1100
8	SCISTOR	HeteroSSL-Fus	1	440M	(1)(3)(4)(5)	-	0.7137 / 0.0314 / 0.3010 / 0.1594
9	JAIST_2	EAT-XLSR-MTL	2	523.09 M	(6)	0.6274 / 0.0977 / 0.2349 / 0.1470	0.7124 / 0.0116 / 0.2604 / 0.1703
10	XHU	CAFM-MTL	2	-	-	-	0.7056 / 0.0172 / 0.1053 / 0.1228
11	JAIST_3	XLSR+BEATs+MH [19]	1	398.140M	(3)(5)(7)(8)	0.7011 / 0.0299 / 0.3140 / 0.1654	0.7019 / 0.0259 / 0.3298 / 0.1883
12	NBU	Feature Decomp.	1	654.00M	-	0.5962 / 0.0179 / 0.4489 / 0.4568	0.6977 / 0.0172 / 0.2307 / 0.2853
13	IITJ	CompMulTask	1	957M	(9)(10)(11)(12)	0.6828 / 0.0632 / 0.1599 / 0.1155	0.6840 / 0.0644 / 0.1587 / 0.1126
-	Baseline	Separation + AASIST	1	957.85M	-	0.6224 / 0.0174 / 0.1993 / 0.4336	0.6327 / 0.0173 / 0.1978 / 0.4279

Data Aug. Methods:
(1)Rawboost; (2)Loudness Aug.; (3)Codec Aug.; (4)Volume Perturbation; (5)Additive Noise; (6)SpecAugment; (7)Mixup; (8)Temporal aug.; (9)Random Cropping; (10)Zero-padding; (11)Mini-batch Shuffling; (12)Class-balanced

fused and mapped to five-class predictions. The separation network operates in the STFT domain, where the speech component is estimated via a learned mask and reconstructed using inverse STFT. The environmental sound is derived from the residual signal with an adaptive soft-masking strategy to suppress speech leakage. The separation network is trained using Cross-Entropy loss on both components.

To ensure spoof-relevant information is preserved, we jointly train the separation and anti-spoofing models. The separated speech and environmental signals are fed into their respective detectors with classification supervision, while a consistency constraint aligns their predictions with those from the reference signals. During inference, the mixture is first classified, then separated into speech and environmental components, which are evaluated independently. Their outputs are combined to produce the final multi-class prediction.

D. Evaluation Criteria

System performance will be evaluated using the **Overall Macro-F1 score** across all five target classes, as shown in (1). The **Macro-F1 score** is computed as the arithmetic mean of the per-class F1 scores. This ensures all classes contribute equally to the final evaluation, regardless of sample imbalance.

$$\text{Macro-F1} = \frac{1}{5}\sum_{i=1}^{5} \text{F1}_i, \tag{1}$$

where, the F1_i score is calculated as the harmonic mean of precision P_i and recall R_i for class i:

$$\text{F1}_i = \frac{2 \cdot P_i \cdot R_i}{P_i + R_i}, \quad P_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}, \quad R_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}. \tag{2}$$

TP_i, FP_i, and FN_i denote the number of true positives, false positives, and false negatives for class i, respectively. A higher Macro-F1 score indicates better overall system performance.

In addition, we introduce three auxiliary metrics based on equal error rate (EER): $\text{EER}_{\text{original}}$, $\text{EER}_{\text{speech}}$ and EER_{env}. $\text{EER}_{\text{original}}$ is the EER for distinguishes the `Original` ($i = 0$) class from all other four classes $i = \{1, 2, 3, 4\}$. Similarly, $\text{EER}_{\text{speech}}$ is the EER for detecting spoofed speech component, while EER_{env} is the EER for detecting spoofed environmental component. The three EER metrics are included for diagnostic and analysis purposes only and are **not** considered for the leaderboard ranking.

III. RESULTS AND DISCUSSIONS

We received 94 registrations from 16 countries. The initial leaderboard contained additional submissions. After verifying the required metadata and challenge submission requirements, 13 teams were retained in the final leaderboard for the analysis. The challenge consists of two phases: a *Preparation Phase* using the evaluation set, and a *Final Ranking Phase* using the test set. The final rankings are determined based on the F1-score in the Final Ranking Phase as present in Table IV.

A. Overall System Trends

The results reveal that the best systems do not rely on one backbone. They combine different models and split the task into clear sub-problems. Top teams often use larger models, but a bigger size does not always mean better results. Team No.1 ranks first with a 6.56B ensemble. Team No.2 ranks second with a much smaller 540.81M cascaded system. This means model complementarity and good decision design matter more than parameter count alone. Efficiency is also important. Many teams improved the baseline while reduced the number of parameters. They saved computation and memory while keeping better performance. This shows better architecture design can outperform simple model scaling.

B. Cross-Domain SSL Backbones

Another clear trend is the use of cross-domain self-supervised learning (SSL) backbones. Besides common speech SSL encoders like XLS-R [20], top systems use newer models such as EAT [21], SSLAM [22], Dasheng [23], and DF-Arena [24]. These models provide different strengths. Speech-focused encoders capture phonetic and prosodic artifacts. Audio/event-focused encoders are better at environmental mismatch and mixture traces. Team No.2 and No.3 use this idea directly by combining EAT/SSLAM-type branches with XLS-R branches. Team No.5 combines DF-Arena with XLSR-Mamba [25], SLS [26], and TCM-ADD [27], then applies component-wise fusion and post-hoc calibration.

C. Ensemble and Fusion Strategies

We also observe that ensemble quality matters more than ensemble size. Team No.5 uses 8 checkpoints, but team No.3 with 4 models and team No.2 with 2 models achieve stronger or similar performance. This suggests that a small set of diverse experts with clear roles is better than a large homogeneous ensemble.

D. Data Augmentation

Data augmentation is still a key factor. RawBoost [28] is the most common method in high-ranked systems. It is often combined with codec simulation, additive noise, and volume perturbation. These methods are especially useful for robust environmental spoofing detection.

E. Architectural Diversity

Importantly, gains do not come only from augmentation or large ensembles. From the system architecture perspective, the top teams do not follow one single architectural style. Instead, they cover multiple strong routes: direct end-to-end 5-class modeling (e.g., team No.1), cascaded coarse-to-fine pipelines (e.g., team No.1 and team No.9), multi-backbone direct fusion (e.g., team No.9 and team No.7), component-level multi-model ensembling with post-hoc calibration (e.g., team No.9), dual-branch speech–environment interaction modeling (e.g., team No.4 and team No.10), separation-driven designs (e.g., team No.6, team No.8 and team No.12), and feature-disentanglement-based designs (e.g., team No.11). Although these systems differ in pipeline form, they share a practical direction: they improve how speech and environmental cues are modeled and then combine those cues with stronger fusion or decision strategies, rather than relying on a single fixed backbone or a single fixed processing stage.

Overall, top submissions follow a common recipe. They use modular designs and combine multiple backbones, especially audio SSL models. They split the task into simpler sub-problems with cascaded, multi-branch, or separation-based pipelines. They do not rely on large models or big ensembles. Instead, they use a small number of diverse models with clear roles, and apply effective fusion and calibration. They also improve efficiency by refining baseline structures to reduce computation while keeping strong performance. Together with solid data augmentation, these designs show diversity, structure, and efficiency matter more than simple model scaling.

IV. Conclusions and Future Scope

In this paper, we summarized the ICME ESDD2 Challenge, including the dataset, task description, baseline system, evaluation criteria, and challenge results analysis. The strong performance achieved by the participating teams highlights the progress made in current deepfake detection methods. At the same time, the results reveal remaining challenges in generalization and robustness across diverse sound types and generation settings. In the future, we plan to further expand the dataset with more realistic and diverse deepfake scenarios and encourage the development of more robust and explainable detection approaches.

V. Acknowledgment

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References

[1] Xueping Zhang, Yechen Wang, Linxi Li, Liwei Jin, and Ming Li, "Compspoof: A dataset and joint learning framework for component-level audio anti-spoofing countermeasures," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 18067–18071.

[2] Han Yin, Yang Xiao, Rohan Kumar Das, Jisheng Bai, and Ting Dang, "Environmental sound deepfake detection challenge: An overview," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 21772–21774.

[3] Han Yin, Yang Xiao, Rohan Kumar Das, Jisheng Bai, and Ting Dang, "Esdd 2026: Environmental sound deepfake detection challenge evaluation plan," *arXiv preprint arXiv:2508.04529*, 2025.

[4] Han Yin, Yang Xiao, Rohan Kumar Das, Jisheng Bai, and Ting Dang, "The first environmental sound deepfake detection challenge: Benchmarking robustness, evaluation, and insights," *arXiv preprint arXiv:2603.04865*, 2026.

[5] Chris Dongjoo Kim, Byeongchang Kim, Hyunmin Lee, and Gunhee Kim, "Audiocaps: Generating captions for audios in the wild," in *The North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 119–132.

[6] Honglie Chen, Weidi Xie, Andrea Vedaldi, and Andrew Zisserman, "Vggsound: A large-scale audio-visual dataset," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 721–725.

[7] Rosana Ardila, Megan Branson, Kelly Davis, Michael Kohler, Josh Meyer, Michael Henretty, Reuben Morais, Lindsay Saunders, Francis Tyers, and Gregor Weber, "Common voice: A massively-multilingual speech corpus," in *Language Resources and Evaluation Conference*, 2020, pp. 4218–4222.

[8] Heiga Zen, Viet Dang, Rob Clark, Yu Zhang, Ron J Weiss, Ye Jia, Zhifeng Chen, and Yonghui Wu, "Libritts: A corpus derived from librispeech for text-to-speech," in *Interspeech*, 2019, pp. 1526–1530.

[9] Jingbei Li, Yi Meng, Chenyi Li, Zhiyong Wu, Helen Meng, Chao Weng, and Dan Su, "Enhancing Speaking Styles in Conversational Text-to-Speech Synthesis with Graph-Based Multi-Modal Context Modeling," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 7917–7921.

[10] Toni Heittola, Annamaria Mesaros, and Tuomas Virtanen, "TAU urban acoustic scenes 2019 open set, development dataset," 2019.

[2] https://ofspectrum.com/

[11] Annamaria Mesaros, Toni Heittola, and Tuomas Virtanen, "Tut database for acoustic scene classification and sound event detection," in *European Signal Processing Conference (EUSIPCO)*, 2016, pp. 1128–1132.

[12] Annamaria Mesaros, Toni Heittola, and Tuomas Virtanen, "Tut sound events 2017, development dataset," 2017.

[13] Annamaria Mesaros, Toni Heittola, and Tuomas Virtanen, "Tut sound events 2017, evaluation dataset," 2017.

[14] Justin Salamon, Christopher Jacoby, and Juan Pablo Bello, "A dataset and taxonomy for urban sound research," in *ACM International Conference on Multimedia (ACM MM)*, 2014, pp. 1041–1044.

[15] Han Yin, Yang Xiao, Rohan Kumar Das, Jisheng Bai, Haohe Liu, Wenwu Wang, and Mark D Plumbley, "Envsdd: Benchmarking environmental sound deepfake detection," in *Interspeech*, 2025, pp. 201–205.

[16] Yuxi Wang, Yikang Wang, Qishan Zhang, Hiromitsu Nishizaki, and Ming Li, "VCapAV: A Video-Caption Based Audio-Visual Deepfake Detection Dataset," in *Interspeech*, 2025, pp. 3908–3912.

[17] Xin Wang, Héctor Delgado, Hemlata Tak, Jee-weon Jung, Hye-jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi H Kinnunen, et al., "Asvspoof 5: crowdsourced speech data, deepfakes, and adversarial attacks at scale," in *ASVspoof*, 2024, pp. 1–8.

[18] Nicolas M Müller, Piotr Kawa, Wei Herng Choong, Edresson Casanova, Eren Gölge, Thorsten Müller, Piotr Syga, Philip Sperl, and Konstantin Böttinger, "Mlaad: The multi-language audio anti-spoofing dataset," in *The International Joint Conference on Neural Networks (IJCNN)*, 2024, pp. 1–7.

[19] Khalid Zaman, Qixuan Huang, Muhammad Uzair, and Masashi Unoki, "Deepfake audio detection using self-supervised fusion representations," *arXiv preprint arXiv:2605.03420*, 2026.

[20] Arun Babu, Changhan Wang, Andros Tjandra, Kushal Lakhotia, Qiantong Xu, Naman Goyal, Kritika Singh, Patrick von Platen, Yatharth Saraf, Juan Pino, et al., "Xls-r: Self-supervised cross-lingual speech representation learning at scale," in *Interspeech*, 2022, pp. 2278–2282.

[21] Wenxi Chen, Yuzhe Liang, Ziyang Ma, Zhisheng Zheng, and Xie Chen, "Eat: self-supervised pre-training with efficient audio transformer," in *The International Joint Conference on Artificial Intelligence (IJCAI)*, 2024, pp. 3807–3815.

[22] Tony Alex, Sara Atito Ali Ahmed, Armin Mustafa, Muhammad Awais, and Philip JB Jackson, "Sslam: Enhancing self-supervised models with audio mixtures for polyphonic soundscapes," in *The International Conference on Learning Representations-Proceedings (ICLR)*, 2025, vol. 2025, pp. 22608–22626.

[23] Heinrich Dinkel, Zhiyong Yan, Yongqing Wang, Junbo Zhang, Yujun Wang, and Bin Wang, "Scaling up masked audio encoder learning for general audio classification," in *Interspeech*, 2024, pp. 547–551.

[24] Ajinkya Kulkarni, Sandipana Dowerah, Atharva Kulkarni, Tanel Alumäe, and Mathew Magimai Doss, "Do compact ssl backbones matter for audio deepfake detection? a controlled study with raptor," *arXiv preprint arXiv:2603.06164*, 2026.

[25] Yang Xiao and Rohan Kumar Das, "Xlsr-mamba: A dual-column bidirectional state space model for spoofing attack detection," in *IEEE Signal Processing Letters*, 2025, vol. 32, pp. 1276–1280.

[26] Qishan Zhang, Shuangbing Wen, and Tao Hu, "Audio deepfake detection with self-supervised xls-r and sls classifier," in *ACM International Conference on Multimedia (ACM MM)*, 2024, pp. 6765–6773.

[27] Duc-Tuan Truong, Ruijie Tao, Tuan Nguyen, Hieu-Thi Luong, Kong Aik Lee, and Eng Siong Chng, "Temporal-channel modeling in multi-head self-attention for synthetic speech detection," in *Interspeech*, 2024, pp. 537–541.

[28] Hemlata Tak, Madhu Kamble, Jose Patino, Massimiliano Todisco, and Nicholas Evans, "Rawboost: A raw data boosting and augmentation method applied to automatic speaker verification anti-spoofing," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 6382–6386.